

Platform Time Synchronization Improvement Plan

Executive decision for high-speed ping-pong operation

Recommendation: replace the stock ntp2rtc system-clock path with governed chrony discipline, while preserving the working HDU-to-PHC, inter-board PTP, and sensor synchronization chain.

Executive summary: benefit versus operational risk

Key improvement benefit	Risk if A3 plays using the stock clock stack
The running HDU clock is continuously disciplined using fractional NTP measurements and delay-aware filtering.	ntp2rtc updates the RTC but does not discipline the running clock; error can grow between boots and during long sessions.
A 100 ppm runtime slew cap and no continuous makestep protect the distributed A3 time rate.	An uncontrolled step or fast correction can propagate through phc2sys/PTP, disturbing board, sensor, recorder, and controller timelines.
Explicit UTC and internal-sync gates expose whether timestamps are trustworthy now.	ROS data may look valid while clock offset, age, or ordering is wrong; stale or out-of-sequence observations may go undetected.
Supervised re-step provides bounded recovery from a bad RTC or long offline boot.	Slew-only recovery can take hours or days; without a runbook the robot may remain unusable or be operated before time is qualified.
Motive remains data-only, so mocap availability cannot alter A3 clocks.	Assuming common NTP servers explicitly synchronize Motive capture time to A3 can hide Wi-Fi delay and timestamp-domain error.

Why this matters during ping-pong

- Ball interception and racket contact depend on consistent measurement age, ordering, and latency across vision/mocap, robot state, and control outputs.
- Clock-domain error can make observations appear earlier or later than they are, bias velocity prediction, and make strike or balance corrections late, inconsistent, or difficult to diagnose.
- This is a credible control-risk pathway, not proof that clock synchronization caused a specific balance loss. Confirm causality with synchronized timestamp traces, controller deadlines, and fall-event logs.

What changes and what does not

Changes	Protected unchanged behavior
chrony owns HDU CLOCK_REALTIME; approved NTP sources; monitored holdover; supervised bootstrap/re-step.	phc2sys direction, active ptp4l E2E behavior, board sync, sensor sync, and application timestamp contracts.

Operational rule: do not permit motion unless both UTC qualification and internal board/sensor synchronization pass. Mocap quality is a separate task-level gate.

1. Design Goals, Scope, and Invariants

Goals

- Continuously discipline the HDU system clock with chrony and expose accuracy, uncertainty, and source health.
- Preserve A3 inter-board, PHC/PTP, sensor, and application timestamp behavior.
- Keep system-clock operation independent of Motive, NatNet, and venue mocap availability.

Scope and non-goals

Domain	In scope	Out of scope
HDU clock	NTP governance, chrony, RTC, bootstrap qualification	Motive host time and capture-time estimation
Internal A3	Preserve phc2sys, ptp4l, board and sensor synchronization	Unvalidated PTP redesign
ROS 2/NatNet	Timestamp-domain boundary and fault isolation	Setting platform time from mocap

Invariants

ID	Requirement
I-1	chrony is the only runtime owner of HDU CLOCK_REALTIME.
I-2	No continuous makestep; steps occur only before protected services or during supervised maintenance.
I-3	Preserve CLOCK_REALTIME -> eth_hdu PHC -> internal-board direction.
I-4	Continuous added slew is capped at 100 ppm.
I-5	Every required board and safety-critical sensor passes pre/post comparison.
I-6	Motive/NatNet cannot control chrony, PHC, PTP, RTC, or clock_settime.
I-7	No production deployment before evidence and owner sign-off.

2. Verified Live A3 Baseline

Read-only inspection on 2026-08-04; no live configuration, service, or clock was changed.

Active ownership path

- HDU runs Debian 12 on aarch64.
- agibot_pm starts /opt/agibot/entry/system/timesync.sh; this is the production owner path.
- The script disables chrony and other time services, sets NTP off, starts ntp2rtc.py, then launches ptp4l and phc2sys.

```
ptp4l -i eth_hdu -2 -E -m  
phc2sys -s CLOCK_REALTIME -c eth_hdu -w -O 0 -S 10 -m
```

Second path and configuration discrepancy

- agibot_timesync.service exists but is disabled/inactive; enabling it would create a second path.

Item	Observed	Decision
Active ptp4l	-E, E2E; no -f	Preserve exactly in Phase 1.
/etc/timesync/ptp_master.conf	delay_mechanism P2P	Inactive; do not enable in this change.
phc2sys	CLOCK_REALTIME source, eth_hdu sink	Preserve direction and arguments.

Evidence and unknowns

- A short phc2sys log sample showed approximately -143 ns to +157 ns; this is a short observation, not a long-term guarantee.
- Per-sensor mechanisms and limits remain a mandatory owner inventory.

Phase 1 replaces only the HDU NTP/system-clock owner; it does not redesign the working internal time chain.

3. Required Platform Time Architecture

Permitted control direction

Approved NTP sources -> chrony -> HDU CLOCK_REALTIME
-> phc2sys -> eth_hdu PHC -> ptp4l -> A3 boards / MDU
-> existing sensor synchronization and timestamp conversion

Layer ownership

Layer	Owner	Contract
UTC reference	Approved NTP set	One documented leap policy; no DHCP injection.
HDU clock	chrony.service	Slew-only, maxslewrate 100, no runtime step.
HDU PHC	Vendor phc2sys	CLOCK_REALTIME source; eth_hdu sink.
Boards	Vendor ptp4l and clients	Preserve active E2E mode, domain, roles, and lock.
Sensors	Firmware/drivers/current conversion	Monotonic, named domain, owner limits.
Applications	ROS 2/robot software	Consume declared domains; no clock ownership.

Motive is a separate data path

Motive Windows -> NatNet -> ROS 2 adapter -> observations
NO path to chrony / CLOCK_REALTIME / PHC / ptp4l / sensor clocks

- Capture-time mapping is optional application logic carrying age and uncertainty; it never feeds platform time.

Separate health conclusions

State	Meaning	Does not prove
UTC-qualified	HDU meets external accuracy/uncertainty gates.	Internal board/sensor coherence.
Internally synchronized	PHC/PTP/boards/sensors meet relative-time gates.	Absolute UTC correctness.

4. Board and Sensor Synchronization Preservation

The existing fact that A3 boards and sensors are synchronized is a protected platform contract, not an implementation to replace.

Mandatory inventory

Object/family	Native clock and timestamp point	Conversion/sync path	Owner limits and failure action
HDU/MDU/boards	PHC, system clock, or device counter	PTP role, offset, domain	offset, jitter, lock, role change
Cameras/vision	Exposure/frame/driver timestamp	Trigger, PTP, or driver mapping	age, drops, reset, alignment
IMU	Sample counter and driver stamp	Device sync or host mapping	monotonicity, drift, jitter
Force/contact	Sample/bus time	Firmware or driver conversion	cycle alignment, timeout
Joint/motor feedback	Board timestamp and bus cycle	Internal board domain	period, offset, duplicate/backward

Required evidence

- Capture at least 30 minutes before and after under equivalent load; report median, p95, p99, maximum offset, jitter, dropout, role changes, and discontinuities.
- Test cold boot, NTP loss/return, chronyd failure, client disconnect, sensor reset, and supervised re-step.

Blocking failures

- Incomplete inventory or unsigned sensor limits.
- UTC passes but internal synchronization fails: motion prohibited.
- Internal synchronization passes but UTC fails: explicit holdover/unqualified state.

Acceptance requires both upstream accuracy and downstream board/sensor coherence.

5. NTP Source Governance and Regional Profiles

China is the default. The operator may switch among the approved china, us, and europe profiles only while robot applications are stopped. Exactly one region is loaded; profiles are never mixed.

Approved official endpoints

Region	Three endpoints	Official authority/source
China■■■■	ntp1.aliyun.com ntp2.aliyun.com ntp3.aliyun.com	Alibaba Cloud public NTP; IPv4 only
US	time-a-g.nist.gov time-b-g.nist.gov time-c-g.nist.gov	NIST Internet Time Service
Europe	ptbtime1.ptb.de ptbtime2.ptb.de ptbtime3.ptb.de	PTB legal-time service

Region selection and operating rules

```
sudo systemctl stop agibot_pm agibot_top agibot_ui
sudo /usr/local/sbin/agibot-time-region china # default
sudo /usr/local/sbin/agibot-time-region us
sudo /usr/local/sbin/agibot-time-region europe
sudo /usr/local/sbin/agibot-time-region --show
chronyc tracking
chronyc sources -v
```

- The selector atomically changes /etc/agibot-time/current and reloads sources with chronyc; UTC and internal synchronization must requalify after a switch.
- Public NTP is unauthenticated. 'Approved' does not remove the network trust boundary, and the profiles do not use chrony's trust option.
- Continuous and bootstrap configs omit leapsetz; leap behavior follows the single active profile. Revalidate provider policy before a scheduled leap event.

Official source references

Alibaba: <https://help.aliyun.com/en/ecs/user-guide/alibaba-cloud-ntp-server/>
NIST: <https://tf.nist.gov/tf-cgi/servers.cgi/en-en/>
PTB: https://www.ptb.de/cms/fileadmin/internet/fachabteilungen/abteilung_8/8.5_metrologische_informationstechnik/8.51/PTB-8.51-MB09-ZS-EN-V09.pdf

6. Continuous Chrony Configuration

/etc/chrony/chrony.conf - single-region profile

```
sourcedir /etc/chrony/sources.d
keyfile /etc/chrony/chrony.keys
driftfile /var/lib/chrony/chrony.drift
ntsdumpdir /var/lib/chrony
logdir /var/log/chrony
rtcsync

minsources 2
maxupdateskew 100.0
maxslewrate 100
corrtimeratio 3
logchange 0.1
log tracking statistics

# Leap handling follows the single selected regional source profile.
# Absent by design: makestep, allow, local, and /run/chrony-dhcp.
```

Controls

Directive	Purpose for A3
sourcedir /etc/chrony/sources.d	Loads only the stable hope-approved.sources link and supports chronyc reload sources.
minsources 2	One reachable source cannot control the clock alone.
maxslewrate 100	Caps runtime added rate so phc2sys/PTP cannot distribute a large rate error.
corrtimeratio 3	Makes correction-time behavior explicit.
■ makestep	Protects live board, sensor, recorder, and control timelines.
■ leapsectz	Avoids an extra assumption about regional-provider leap handling.

Service supervision

```
[Unit]
After=agibot-clock-bootstrap.service
StartLimitIntervalSec=60
StartLimitBurst=3

[Service]
Restart=on-failure
RestartSec=5
```

- Any additional *.sources file beside hope-approved.sources makes preflight fail.

7. Replace ntp2rtc Without Replacing Internal Sync

Appendix A.6 and `agibot/ntp_sync/opt/agibot/entry/system/timesync.sh` contain the exact reviewable replacement. The package file is canonical; do not reconstruct it from PDF text.

Verified active owner

- `/opt/agibot/entry/system/timesync.sh` is launched in the background by `agibot_pm`'s `run_agibot.sh`.
- The `systemd` drop-in synchronously executes `timesync.sh --preflight` so a background launcher failure cannot bypass startup gating.

Changes implemented by the complete replacement

Removed or changed	Preserved or added
No <code>chrony</code> stop/disable; remove <code>timedatectl set-ntp false</code> and <code>ntp2rtc</code> process logic.	Preserve log rotation and <code>ptp4l/phc2sys</code> process management.
Do not use the inactive P2P configuration file.	Preserve <code>ptp4l -2 -E</code> and <code>CLOCK_REALTIME -> eth_hdu phc2sys</code> .
UTC-unqualified operation is denied by default.	Add regional-profile, selected-source, and signed holdover checks.
<code>LOWER_UP</code> alone is not sufficient readiness.	Require <code>ethtool -T</code> PTP hardware clock plus hardware TX/RX/raw-clock capabilities.

Deployment metadata and canonical location

- Back up `timesync.sh` with `cp -a` and record `stat/SHA-256` before replacement; installation preserves the observed numeric owner, group, and mode.
- Never assume `agi:agi` across the A3 fleet, and never reapply automatically after an unknown vendor update.
- The only canonical engineering package is `agibot/ntp_sync/`. Historical generated package copies are superseded.

Phase 1A installs and validates with services disabled; it does not authorize robot motion.

8. Boot Bootstrap and Qualification Markers

Bootstrap configuration

```
sourcedir /etc/chrony/sources.d
driftfile /var/lib/chrony/chrony.drift
makestep 0.001 1
maxsamples 1
minsources 2
maxupdateskew 100.0
logdir /var/log/chrony
logchange 0.1
```

Regional and file qualification

- The A.1 script accepts only a stable link resolving into china/us/europe and requires exactly one *.sources file in sources.d.
- The active profile is root-owned, not group/world writable, and contains exactly three non-placeholder server entries.

Bounded execution and markers

```
# /etc/default/agibot-clock-bootstrap
BOOTSTRAP_TIMEOUT_S=30          # measured venue value; 5..150

install -d -o root -g root -m 0755 /run/agibot-time
chronyd -p -f /etc/chrony/chrony-bootstrap.conf
timeout --signal=TERM "$((BOOTSTRAP_TIMEOUT_S + 5))s" \
  chronyd -q -t "$BOOTSTRAP_TIMEOUT_S" \
  -f /etc/chrony/chrony-bootstrap.conf
```

Marker	Condition and meaning
bootstrap-qualified	Created after chronyd -q succeeds and before hwclock; RTC failure does not remove it.
rtc-updated	Created only after hwclock --systohc --utc succeeds.
runtime qualification	Continuous chrony gates currently pass.
internal-sync qualification	Board and sensor gates currently pass.

The complete A.1 script is on page 16 and in the package; do not assemble it from this page.

9. Systemd Ordering and Supervised Re-step

Boot ordering

```
[Unit]
Description=Qualify A3 system time before the distributed robot time domain
Wants=network-online.target
After=network-online.target
Before=chrony.service agibot_roudi.service agibot_pm.service
Before=agibot_top.service agibot_ui.service

[Service]
Type=oneshot
ExecStart=/usr/local/sbin/agibot-clock-bootstrap
TimeoutStartSec=180s
RemainAfterExit=yes

[Install]
WantedBy=multi-user.target
```

Synchronous agibot_pm gate

```
[Unit]
After=chrony.service

[Service]
ExecStartPre=/opt/agibot/entry/system/timesync.sh --preflight
```

- After=chrony.service orders only; it adds no Wants= or Requires= dependency.
- ExecStartPre synchronously runs --preflight from the complete timesync.sh, compensating for run_agibot.sh launching its normal path in the background.
- By default preflight requires active chrony, a valid profile, Leap status Normal, and a ^* selected source. AGIBOT_ALLOW_UNQUALIFIED_TIME=1 requires a signed holdover procedure.

Region switch and re-step

- The selector refuses a switch while agibot_pm is active; afterward it reloads/refreshes sources and all qualification gates rerun.
- Large-offset re-step still stops applications, chrony, and downstream distribution, requires two-person approval, and requalifies every layer before motion.

A background timesync.sh failure is not a safety gate; only the synchronous systemd ExecStartPre exit status blocks agibot_pm.

10. End-to-End Acceptance Criteria

Layered acceptance

Layer	Pass criteria
chrony	Approved sources; maxslewrate 100; corrtimeratio 3; no runtime step or DHCP source.
HDU clock	Selected source, Leap Normal, correction/delay/dispersion/skew within limits.
Clock rate	No step; CLOCK_REALTIME versus MONOTONIC_RAW within cap plus measured tolerance.
PHC	phc2sys direction unchanged; distribution no worse than baseline.
Boards	One master; expected clients locked; no role/domain/mechanism regression.
Sensors	Monotonic timestamps in intended domain; offset/jitter/age/dropout pass.
Applications	No backward/duplicate stamps, timeout storms, corruption, or instability.
Mocap independence	Motive disconnect/restart has zero effect on A3 clock and sync processes.

Required comparison

- At least 30 minutes pre/post under equivalent load; report median, p95, p99, maximum offset, jitter, dropouts, role changes, and discontinuities.

Provisional gates to replace with owner limits

Metric	Provisional rule
remaining correction	10 ms example only.
chrony skew	5 ppm example only; characterize on venue Wi-Fi.
phc2sys offset	Post-change p99/max do not regress.
board offset	Use vendor/A3 limit per client.
sensor alignment	Use fusion/control limit per family.

No single synchronized flag is sufficient; prove upstream accuracy and downstream coherence separately.

11. Recovery Timing and Fault Injection

100 ppm recovery math

A 100 ppm slew corrects at most 0.0001 seconds per elapsed second. It protects the distributed rate but makes large correction slow.

Offset	Best-case time	Action
10 ms	100 s	Remain gated.
0.3 s	50 min	Operational re-step decision.
2 s	5 h 33 min	Normally supervised re-step.
30 s	83 h 20 min (3.47 d)	Supervised re-step required.

Fault matrix

Scenario	Required result
NTP unavailable at boot	Bounded delay; holdover only; UTC unqualified.
Silent UDP/123 drop	Exit at measured timeout.
NTP returns with large offset	No runtime step; 100 ppm cap; recovery estimate shown.
chronyd crashes	Restart within 5 s; repeated failures alarm.
RTC write fails	Runtime marker remains; RTC marker absent.
DHCP offers NTP	No source added.
Motive restarts	No A3 clock or sync change.
PTP client disconnects	Affected board alarm; no unintended GM takeover.
Sensor timestamp resets	Discontinuity detected; approved inhibit/degrade action.
Supervised re-step	All layers stop and requalify before motion.

Offline boot timing

- Measure five online boots, five immediate DNS failures, and five silent firewall drops before accepting the timeout.
- The supported venue parameter is 5..150 seconds with a 180-second unit ceiling.

12. Monitoring, Rollout, and Rollback

Continuous monitoring

Domain	Monitor
NTP/HDU	tracking, sources, sourcestats, restart count, identity, correction, skew, dispersion
PHC	phc2sys offset/frequency/delay distribution and arguments
PTP boards	roles, changes, offset, mechanism, domain, reachability, duplicate owners
Sensors	native state, counters, monotonicity, offset/jitter/age, drops, resets
Applications	backward stamps, timeout bursts, ordering, control anomalies
Independence	Motive changes produce zero platform-clock events

Rollout

Phase	Change	Exit gate
0	Inventory and baseline	All mechanisms and limits recorded.
1A	Install disabled files	No active owner change.
1B	Bench migration	Layered tests and fault matrix pass.
1C	Venue Wi-Fi rehearsal	NTP policy, timeout, holdover approved.
1D	Production rollout in physical safe state	Cold boots, soak, sign-off complete.
2	Optional PTP service refactor	Separate board/sensor compatibility change.

Rollback trigger and procedure

- Any board/sensor regression, rate violation, discontinuity, duplicate owner, role/direction change, instability, or source-governance failure.

```
sudo systemctl stop agibot_pm agibot_roudi agibot_top agibot_ui
sudo systemctl stop chrony agibot-clock-bootstrap
sudo systemctl disable chrony agibot-clock-bootstrap
# Restore signed vendor script, units, and configuration.
sudo systemctl daemon-reload
sudo systemctl start agibot_roudi agibot_top agibot_ui agibot_pm
```

Rollback restores prior behavior but not accepted continuous UTC discipline; mark external-time correlation unqualified.

13. Motive and ROS 2 Boundary

Motive is an external observation system. It may affect behavior through data, but cannot affect A3 clock discipline or the internal board/sensor time domain.

Required separation

Question	Answer
Must Windows be PTP GM?	No. A3 does not follow Motive by PTP.
Must A3 and Windows share NTP?	No. It may aid diagnosis but is not an A3 dependency.
Can NatNet discipline A3?	No clock_settime, chrony, PTP, PHC, or RTC control from mocap.
How should ROS stamp data?	Use a documented consumer domain; preserve native Motive metadata.
Capture-time mapping allowed?	Yes as isolated application estimate with age/uncertainty.
Does mocap quality gate system time?	No. It may gate the ping-pong task only.

Message semantics

Field	Meaning
header.stamp	Documented A3/relay-domain time.
motive_frame_number	Native identity for loss/reordering.
motive_native_timestamp	Preserved native counter; not claimed as A3 time.
receive_timestamp	Adapter receive time for age/jitter.
capture_time_estimate	Optional mapped estimate with age/uncertainty.

Failure isolation

- Disconnect Motive, stop/restart it and NatNet, inject malformed stamps and delay/reordering.
- Prove chronyd, CLOCK_REALTIME rate, phc2sys, ptp4l, board lock, and sensor synchronization remain unchanged.
- Mocap-specific capture-time estimation belongs to HOPE integration, not A3 system-clock service.

14. Review Disposition, References, and Sign-off

v1.5 review disposition

Item	Disposition	Result
A.6 script drift	Resolved - blocking	A.6 is regenerated from agibot/ntp_sync and includes ethtool PHY readiness.
A.8 backup metadata	Resolved - blocking	cp -a preserves ownership; chrony.conf, package versions, and enablement are captured.
A.9 rollback assumptions	Resolved - blocking	Rollback restores captured metadata/config and removes introduced files.
Canonical package ambiguity	Resolved	agibot/ntp_sync is the only canonical engineering source.
Package presence assumptions	Resolved	The runbook detects dependencies and installs only genuinely missing packages.
UTC-unqualified holdover	Resolved	README section 6 defines two-person enable, marker, monitoring, and fail-closed exit.
10 ms / 5 ppm examples	Open gate	Owner-signed replacements remain required before Phase 1B.

Official references

- [1] Alibaba Cloud public NTP: <https://help.aliyun.com/en/ecs/user-guide/alibaba-cloud-ntp-server/>
[2] NIST Internet Time Service: <https://tf.nist.gov/tf-cgi/servers.cgi/en-en/>
[3] PTB NTP/NTS legal time: https://www.ptb.de/cms/fileadmin/internet/fachabteilungen/abteilung_8/8.5_metrologische_informationstechnik/8.51/PTB-8.51-MB09-ZS-EN-V09.pdf
[4] chronyc: <https://chrony-project.org/doc/4.4/chronyc.html>
[5] chrony 4.3 configuration: <https://chrony-project.org/doc/4.3/chrony.conf.html>
[6] LinuxPTP: <https://www.linuxptp.org/documentation/>

Final authorization

Role	Name/signature	Date	Decision
Robot controls			GO / NO-GO
Board/firmware			GO / NO-GO
Sensor integration			GO / NO-GO
Platform			GO / NO-GO
Event operations			GO / NO-GO

Before Phase 1A, verify the package manifest and current A.6 source. Before Phase 1B, complete sensor inventory and owner limits.

Appendix A. Complete Deployment Artifacts

A.1 Complete root-owned bootstrap script

Identical to agibot/ntp_sync/usr/local/sbin/agibot-clock-bootstrap.

```
#!/bin/bash

set -Eeuo pipefail
umask 022
PATH=/usr/sbin:/usr/bin:/sbin:/bin

readonly CONF=/etc/chrony/chrony-bootstrap.conf
readonly SOURCES=/etc/chrony/sources.d/hope-approved.sources
readonly REGION_ROOT=/etc/agibot-time/regions
readonly DEFAULTS=/etc/default/agibot-clock-bootstrap
readonly STATE_DIR=/run/agibot-time
readonly QUALIFIED="${STATE_DIR}/bootstrap-qualified"
readonly RTC_UPDATED="${STATE_DIR}/rtc-updated"

log() { logger -t agibot-clock-bootstrap -- "$*"; printf '%s\n' "$*"; }
fail() { log "ERROR: $*"; exit 1; }

validate_file() {
    local file=$1 perm
    [[ -f "$file" ]] || fail "missing regular file: $file"
    [[ $(stat -Lc %u "$file") -eq 0 ]] || fail "not root-owned: $file"
    perm=$(stat -Lc %a "$file")
    (( ${perm} & 8#022 == 0 )) || fail "group/world writable: $file"
}

[[ $EUID -eq 0 ]] || fail "must run as root"
install -d -o root -g root -m 0755 "$STATE_DIR"
rm -f "$QUALIFIED" "$RTC_UPDATED"

validate_file "$CONF"
validate_file "$SOURCES"
target=$(readlink -f "$SOURCES")
case "$target" in
    "$REGION_ROOT"/china/chrony.sources|\
    "$REGION_ROOT"/us/chrony.sources|\
    "$REGION_ROOT"/europe/chrony.sources) ;;
    *) fail "unapproved regional source target: $target" ;;
esac

mapfile -t source_files < <(compgen -G '/etc/chrony/sources.d/*.sources' || true)
[[ ${#source_files[@]} -eq 1 && "${source_files[0]}" == "$SOURCES" ]] ||
    fail "/etc/chrony/sources.d must contain only hope-approved.sources"
[[ $(grep -Ec '^[[:space:]]*server[[:space:]]+' "$SOURCES" || true) -eq 3 ]] ||
    fail "selected regional profile must contain exactly three servers"
grep -Eq 'site\,example|REPLACE_ME' "$SOURCES" && fail "placeholder NTP source"

if [[ -e "$DEFAULTS" ]]; then
    validate_file "$DEFAULTS"
    # shellcheck source=/etc/default/agibot-clock-bootstrap
    source "$DEFAULTS"
fi
: "${BOOTSTRAP_TIMEOUT_S:=30}"
[[ $BOOTSTRAP_TIMEOUT_S =~ ^[0-9]+$ ]] || fail "timeout is not an integer"
(( BOOTSTRAP_TIMEOUT_S >= 5 && BOOTSTRAP_TIMEOUT_S <= 150 )) ||
    fail "timeout must be within 5..150 seconds"

systemctl is-active --quiet chrony.service && fail "continuous chrony is already active"
chronyd -p -f "$CONF" >/dev/null || fail "bootstrap configuration rejected"
outer_timeout=$((BOOTSTRAP_TIMEOUT_S + 5))
if timeout --signal=TERM "${outer_timeout}s" \
    chronyd -q -t "$BOOTSTRAP_TIMEOUT_S" -f "$CONF"; then
    touch "$QUALIFIED"
else
    fail "approved network time was not applied"
fi

if command -v hwclock >/dev/null 2>&1 && hwclock --systohc --utc; then
    touch "$RTC_UPDATED"
else
    log "WARNING: RTC update failed; runtime qualification remains valid"
fi
log "bootstrap-qualified; rtc_updated=${[[ -e $RTC_UPDATED ]] && echo yes || echo no}"
```

Appendix A. Complete Deployment Artifacts

A.2 Bootstrap service

```
[Unit]
Description=Qualify A3 system time before the distributed robot time domain
Wants=network-online.target
After=network-online.target
Before=chrony.service agibot_roudi.service agibot_pm.service
Before=agibot_top.service agibot_ui.service

[Service]
Type=oneshot
ExecStart=/usr/local/sbin/agibot-clock-bootstrap
TimeoutStartSec=180s
RemainAfterExit=yes

[Install]
WantedBy=multi-user.target
```

A.3 Continuous chrony supervision

```
[Unit]
After=agibot-clock-bootstrap.service
StartLimitIntervalSec=60
StartLimitBurst=3

[Service]
Restart=on-failure
RestartSec=5
```

A.4 agibot_pm ordering and synchronous gate

```
[Unit]
After=chrony.service

[Service]
ExecStartPre=/opt/agibot/entry/system/timesync.sh --preflight
```

A.5 Package integrity and metadata

- All canonical engineering files are under agibot/ntp_sync; run sha256sum -c MANIFEST.sha256 before installation.
- New config/unit files are root-owned. The vendor timesync.sh replacement preserves the observed numeric owner, group, and mode.
- README.md is the canonical operational procedure for dependency detection, holdover, activation, and rollback.

Do not copy commands from a superseded v1.4 PDF or historical generated package tree.

Appendix A. Complete Deployment Artifacts

A.6 Complete Modified timesync.sh (Part 1 of 4)

The canonical file is agibot/ntp_sync/opt/agibot/entry/system/timesync.sh. The four parts below are generated from that exact file; do not reconstruct deployment content from the PDF.

```
#!/bin/bash

# Reviewed replacement for the vendor A3 time-distribution launcher.
# The system clock is owned by chrony. This script preserves the active
# ptp4l E2E and CLOCK_REALTIME-to-eth_hdu phc2sys behavior.

set -u -o pipefail
export LC_ALL=C

cd -- "$(dirname -- "$0")" || exit 1

LOG_DIR="/agibot/log/tsync"
MAX_LOG_FILES=20
device_name="eth_hdu"
TIMESTAMP=$(date +"%Y%m%d_%H%M%S")
NTP_LOG="$LOG_DIR/ntp_${TIMESTAMP}.log"
PTP4L_LOG="$LOG_DIR/ptp4l_${TIMESTAMP}.log"
PHC2SYS_LOG="$LOG_DIR/phc2sys_${TIMESTAMP}.log"
ACTIVE_SOURCES="/etc/chrony/sources.d/hope-approved.sources"
REGION_ROOT="/etc/agibot-time/regions"
POLICY_FILE="/etc/default/agibot-timesync"
AGIBOT_ALLOW_UNQUALIFIED_TIME=0

mkdir -p "$LOG_DIR"

log() {
    local log_file=$1
    shift
    printf '%s\n' "$@" | tee -a "$log_file"
}

rotate_logs() {
    local prefix=$1
    local max_count=$2
    ls -t "$LOG_DIR/${prefix}"_*.log 2>/dev/null |
        sed -e "1,${max_count}d" |
        xargs -r rm
}

validate_root_file() {
    local file=$1
    local perm
    [[ -f "$file" ]] || return 1
    [[ $(stat -Lc %u "$file") -eq 0 ]] || return 1
    perm=$(stat -Lc %a "$file")
    (( (8#$perm & 8#022) == 0 ))
}

validate_region_profile() {
```

Appendix A. Complete Deployment Artifacts

A.6 Complete Modified timesync.sh (Part 2 of 4)

The canonical file is agibot/ntp_sync/opt/agibot/entry/system/timesync.sh. The four parts below are generated from that exact file; do not reconstruct deployment content from the PDF.

```
local target
local source_count
local source_files

[[ -L "$ACTIVE_SOURCES" ]] || {
    log "$NTP_LOG" "ERROR: active regional source link is missing: $ACTIVE_SOURCES"
    return 1
}
target=$(readlink -f "$ACTIVE_SOURCES") || return 1
case "$target" in
    "$REGION_ROOT"/china/chrony.sources\
    "$REGION_ROOT"/us/chrony.sources\
    "$REGION_ROOT"/europe/chrony.sources) ;;
    *)
        log "$NTP_LOG" "ERROR: unapproved regional source target: $target"
        return 1
        ;;
esac
validate_root_file "$target" || {
    log "$NTP_LOG" "ERROR: regional source file ownership or mode is unsafe: $target"
    return 1
}
grep -Eq 'site\\.example|REPLACE_ME' "$target" && {
    log "$NTP_LOG" "ERROR: placeholder NTP source remains in $target"
    return 1
}
source_count=$(grep -Ec '^[[:space:]]*server[[:space:]]+' "$target" || true)
[[ $source_count -eq 3 ]] || {
    log "$NTP_LOG" "ERROR: regional profile must contain exactly three servers: $target"
    return 1
}
mapfile -t source_files < <(compgen -G '/etc/chrony/sources.d/*.sources' || true)
[[ ${#source_files[@]} -eq 1 && "${source_files[0]}" == "$ACTIVE_SOURCES" ]] || {
    log "$NTP_LOG" "ERROR: /etc/chrony/sources.d must contain only hope-approved.sources"
    return 1
}
log "$NTP_LOG" "Approved regional source profile: $target"
}

chrony_preflight() {
    local tracking

    validate_region_profile || return 1
    systemctl is-active --quiet chrony.service || {
        log "$NTP_LOG" "ERROR: chrony.service is not active"
        return 1
    }
    tracking=$(chronyc tracking 2>&1) || {
        log "$NTP_LOG" "ERROR: chronyc tracking failed: $tracking"
    }
}
```

Appendix A. Complete Deployment Artifacts

A.6 Complete Modified timesync.sh (Part 3 of 4)

The canonical file is `agibot/ntp_sync/opt/agibot/entry/system/timesync.sh`. The four parts below are generated from that exact file; do not reconstruct deployment content from the PDF.

```
        return 1
    }
    if grep -q '^Leap status[:space:]*[:space:]*Normal' <<<"Stracking" &&
        chronyc -n sources 2>/dev/null | grep -q '^\\^\\^'; then
        log "SNTP_LOG" "chrony is active and synchronized to a selected NTP source"
        return 0
    fi
    if [[ "$AGIBOT_ALLOW_UNQUALIFIED_TIME" == "1" ]]; then
        log "SNTP_LOG" "WARNING: continuing in explicitly approved UTC-unqualified holdover mode"
        return 0
    fi
    log "SNTP_LOG" "ERROR: UTC is not qualified; set AGIBOT_ALLOW_UNQUALIFIED_TIME=1 only under an approved holdover procedure"
    return 1
}

ensure_killed() {
    local process_name=$1
    local log_file=$2
    while pgrep -x "$process_name" >/dev/null; do
        log "$log_file" "Stopping existing $process_name process..."
        pkill -x "$process_name" || true
        sleep 2
        pgrep -x "$process_name" >/dev/null && pkill -9 -x "$process_name" || true
        sleep 1
    done
    log "$log_file" "$process_name process is stopped."
}

rotate_logs "ntp" "$MAX_LOG_FILES"
rotate_logs "ptp4l" "$MAX_LOG_FILES"
rotate_logs "phc2sys" "$MAX_LOG_FILES"

if [[ -e "$POLICY_FILE" ]]; then
    validate_root_file "$POLICY_FILE" || {
        log "SNTP_LOG" "ERROR: unsafe policy file: $POLICY_FILE"
        exit 1
    }
    # shellcheck source=/etc/default/agibot-timesync
    source "$POLICY_FILE"
fi

# Synchronous systemd ExecStartPre entry point. This is required because the
# vendor run_agibot.sh launches the normal timesync.sh path in the background.
if [[ "${1:-}" == "--preflight" ]]; then
    chrony_preflight
    exit $?
fi

# Stop conflicting vendor services, but never stop or disable chrony.
```

Appendix A. Complete Deployment Artifacts

A.6 Complete Modified timesync.sh (Part 4 of 4)

The canonical file is agibot/ntp_sync/opt/agibot/entry/system/timesync.sh. The four parts below are generated from that exact file; do not reconstruct deployment content from the PDF.

```
SERVICES=("ptp4l" "phc2sys" "agibot_timesync" "ntpd" "ntpsec")
for SERVICE in "${SERVICES[@]}; do
    if systemctl is-active --quiet "$SERVICE"; then
        log "$PTP4L_LOG" "Detected active $SERVICE; stopping and disabling it..."
        systemctl disable --now "$SERVICE"
        log "$PTP4L_LOG" "$SERVICE is stopped and disabled."
    else
        log "$PTP4L_LOG" "$SERVICE is not active."
    fi
done

while true; do
    if ip link show "$device_name" 2>/dev/null | grep -q "LOWER_UP"; then
        TIMESTAMPING_INFO=$(ethtool -T "$device_name" 2>&1) || {
            log "$PTP4L_LOG" "ERROR: ethtool could not query $device_name timestamping: $TIMESTAMPING_INFO"
            exit 1
        }
        grep -Eq 'PTP Hardware Clock:[[:space:]]*[0-9]+' <<<"$TIMESTAMPING_INFO" || {
            log "$PTP4L_LOG" "ERROR: $device_name does not expose a PTP hardware clock"
            exit 1
        }
        for CAPABILITY in hardware-transmit hardware-receive hardware-raw-clock; do
            grep -q "$CAPABILITY" <<<"$TIMESTAMPING_INFO" || {
                log "$PTP4L_LOG" "ERROR: $device_name lacks $CAPABILITY timestamping capability"
                exit 1
            }
        done
        log "$PTP4L_LOG" "Interface $device_name is Link Up with hardware PTP timestamping."
        break
    fi
    log "$PTP4L_LOG" "ERROR: $device_name is not ready; waiting..."
    sleep 2
done

chrony_preflight || exit 1

if pgrep -x ptp4l >/dev/null; then
    ensure_killed "ptp4l" "$PTP4L_LOG"
fi
log "$PTP4L_LOG" "Starting ptp4l master on $device_name in preserved E2E mode..."
nohup ptp4l -i "$device_name" -2 -E -m \
    2>&1 | awk '{print strftime("[%F %T]"), $0}' >> "$PTP4L_LOG" &

if pgrep -x phc2sys >/dev/null; then
    ensure_killed "phc2sys" "$PHC2SYS_LOG"
fi
log "$PHC2SYS_LOG" "Starting phc2sys (CLOCK_REALTIME -> $device_name PHC)..."
nohup phc2sys -s CLOCK_REALTIME -c "$device_name" -w -0 0 -S 10 -m \
    2>&1 | awk '{print strftime("[%F %T]"), $0}' >> "$PHC2SYS_LOG" &
```

Appendix A. Complete Deployment Artifacts

A.7 Three Regional Source Profiles

China is the default. These exact files are included in agibot/ntp_sync.

```
# Region: China (default)
# Authority: Alibaba Cloud public NTP service, official ECS documentation.
# Public service is IPv4-only. Do not add sources from another region.
server ntp1.aliyun.com iburst minpoll 6 maxpoll 9 maxdelay 0.30
server ntp2.aliyun.com iburst minpoll 6 maxpoll 9 maxdelay 0.30
server ntp3.aliyun.com iburst minpoll 6 maxpoll 9 maxdelay 0.30

# Region: United States
# Authority: NIST Internet Time Service. Do not add sources from another region.
server time-a-g.nist.gov iburst minpoll 6 maxpoll 9 maxdelay 0.30
server time-b-g.nist.gov iburst minpoll 6 maxpoll 9 maxdelay 0.30
server time-c-g.nist.gov iburst minpoll 6 maxpoll 9 maxdelay 0.30

# Region: Europe
# Authority: PTB legal-time NTP service. Do not add sources from another region.
server ptbtime1.ptb.de iburst minpoll 6 maxpoll 9 maxdelay 0.30
server ptbtime2.ptb.de iburst minpoll 6 maxpoll 9 maxdelay 0.30
server ptbtime3.ptb.de iburst minpoll 6 maxpoll 9 maxdelay 0.30
```

A.7 Region Selector (Part 1 of 2)

```
#!/bin/bash

set -Eeuo pipefail
export LC_ALL=C

REGION_ROOT="/etc/agibot-time/regions"
CURRENT_LINK="/etc/agibot-time/current"
ACTIVE_LINK="/etc/chrony/sources.d/hope-approved.sources"

fail() {
    logger -t agibot-time-region -- "ERROR: $"
    printf 'ERROR: %s\n' "$@" >&2
    exit 1
}

show_region() {
    local target
    target=$(readlink -f "$CURRENT_LINK" 2>/dev/null || true)
    [[ -n "$target" ]] || fail "no active region"
    basename "$target"
}

if [[ "${1:-}" == "--show" ]]; then
    show_region
    exit 0
fi

[[ $EUID -eq 0 ]] || fail "must run as root"
region=${1:-china}
case "$region" in
    china|us|europe) ;;
    *) fail "region must be china, us, or europe" ;;
esac

systemctl is-active --quiet agibot_pm.service &&
fail "stop agibot_pm.service before switching regions"

target="$REGION_ROOT/$region"
```

Appendix A. Complete Deployment Artifacts

A.7 Region Selector (Part 2 of 2)

```
profile="Target/chrony.sources"
[[ -f "$profile" ]] || fail "missing profile: $profile"
[[ $(stat -Lc %u "$profile") -eq 0 ]] || fail "profile is not root-owned"
perm=$(stat -Lc %a "$profile")
(( ($perm & 8#022) == 0 )) || fail "profile is group/world writable"
[[ $(grep -Ec '^[[:space:]]server[[:space:]]+' "$profile" | true) -eq 3 ]] ||
    fail "profile must contain exactly three servers"
grep -Eq 'site\.example|REPLACE_ME' "$profile" && fail "profile contains a placeholder"

install -d -o root -g root -m 0755 /etc/agibot-time /etc/chrony/sources.d
tmp_link="$CURRENT_LINK.new.$$"
trap 'rm -f "$tmp_link" EXIT'
ln -s "Target" "$tmp_link"
mv -Tf "$tmp_link" "$CURRENT_LINK"
trap - EXIT

if [[ ! -L "$ACTIVE_LINK" ]]; then
    ln -s /etc/agibot-time/current/chrony.sources "$ACTIVE_LINK"
fi
[[ $(readlink "$ACTIVE_LINK") == "/etc/agibot-time/current/chrony.sources" ]] ||
    fail "unexpected active source link: $ACTIVE_LINK"

logger -t agibot-time-region -- "selected region=$region profile=$profile"
printf 'Selected region: %s\n' "$region"

if systemctl is-active --quiet chrony.service; then
    chronyc reload sources
    chronyc online
    chronyc refresh
    chronyc sources -v
fi

printf '%s\n' "UTC is unqualified until chrony and internal board/sensor gates pass."
```

Operator contract

- Switch only while robot applications and downstream distribution are stopped.
- Exactly one .sources file is permitted; requalify UTC, PHC, PTP, boards, and sensors after switching.
- Leap handling follows README section 3.4: Aliyun is treated as provider-managed smear time; NIST/PTB as standard UTC; leapsectz is omitted globally.

Appendix A. Complete Deployment Artifacts

A.8 Metadata-preserving Backup and Baseline Commands (Part 1 of 2)

Generated from README section 2. README.md remains the canonical operational runbook.

```
STAMP=$(date -u +%Y%m%dT%H%M%S)
BACKUP="/var/backups/agibot-time/$STAMP"
echo "BACKUP=$BACKUP"

sudo install -d -o root -g root -m 0700 "$BACKUP/rootfs"
sudo cp -a --parents /opt/agibot/entry/system/timesync.sh "$BACKUP/rootfs"
sudo sha256sum /opt/agibot/entry/system/timesync.sh | \
  sudo tee "$BACKUP/vendor-timesync.sha256"
sudo stat -Lc '%n %U:%G %a %s %y' \
  /opt/agibot/entry/system/timesync.sh | \
  sudo tee "$BACKUP/vendor-timesync.stat"

if sudo test -e /etc/chrony/chrony.conf; then
  sudo cp -a --parents /etc/chrony/chrony.conf "$BACKUP/rootfs"
  sudo sha256sum /etc/chrony/chrony.conf | \
    sudo tee "$BACKUP/chrony-conf.sha256"
  sudo touch "$BACKUP/chrony-conf-present"
else
  sudo touch "$BACKUP/chrony-conf-absent"
fi

dpkg-query -W -f='${Package}\t${Version}\t${db:Status-Abbrev}\n' \
  bash chrony coreutils ethtool findutils grep iproute2 linuxptp mawk \
  procs sed systemd util-linux-extra bsdutils 2>/dev/null | \
  sudo tee "$BACKUP/dependency-packages.txt" || true
apt-cache policy chrony ethtool linuxptp | \
```

Appendix A. Complete Deployment Artifacts

A.8 Metadata-preserving Backup and Baseline Commands (Part 2 of 2)

Generated from README section 2. README.md remains the canonical operational runbook.

```
sudo tee "$BACKUP/dependency-policy.txt" || true
systemctl is-enabled chrony.service 2>&1 | \
sudo tee "$BACKUP/chrony-enabled.txt" || true
systemctl is-active chrony.service 2>&1 | \
sudo tee "$BACKUP/chrony-active.txt" || true
sudo systemctl cat agibot_pm.service | \
sudo tee "$BACKUP/agibot_pm.service.txt"
sudo systemctl list-dependencies agibot_pm.service | \
sudo tee "$BACKUP/agibot_pm.dependencies.txt"
sudo pgrep -a -f 'ntp2rtc|chronyd|ptp4l|phc2sys' | \
sudo tee "$BACKUP/time-processes.txt" || true
sudo timedatectl status | sudo tee "$BACKUP/timedatectl.txt"

if sudo test -d /agibot/log/tsync; then
sudo cp -a /agibot/log/tsync "$BACKUP/tsync-logs"
fi

sudo ln -sfn "$BACKUP" /var/backups/agibot-time/latest
sudo readlink -f /var/backups/agibot-time/latest

sudo test -s "$BACKUP/vendor-timesync.sha256"
sudo test -s "$BACKUP/vendor-timesync.stat"
sudo test -e "$BACKUP/chrony-conf-present" -o \
-e "$BACKUP/chrony-conf-absent"
sudo ls -la "$BACKUP"
```

Appendix A. Complete Deployment Artifacts

A.9 Metadata-preserving Rollback Commands (Part 1 of 2)

Generated from README section 11. Stop on any failed prerequisite or unknown prior state.

```
BACKUP=$(sudo readlink -f /var/backups/agibot-time/latest)
echo "BACKUP=$BACKUP"
sudo test -s "$BACKUP/vendor-timesync.sha256"
sudo test -s "$BACKUP/vendor-timesync.stat"
sudo ls -la "$BACKUP/rootfs/opt/agibot/entry/system/timesync.sh"
sudo test -e "$BACKUP/chrony-conf-present" -o \
-e "$BACKUP/chrony-conf-absent"

sudo systemctl stop agibot_pm agibot_roudi agibot_top agibot_ui || true
sudo pkill -TERM -x phc2sys || true
sudo pkill -TERM -x ptp4l || true
sudo systemctl disable --now chrony.service agibot-clock-bootstrap.service || true

sudo cp -a \
"$BACKUP/rootfs/opt/agibot/entry/system/timesync.sh" \
/opt/agibot/entry/system/timesync.sh

if sudo test -e "$BACKUP/chrony-conf-present"; then
    sudo cp -a "$BACKUP/rootfs/etc/chrony/chrony.conf" \
    /etc/chrony/chrony.conf
elif sudo test -e "$BACKUP/chrony-conf-absent"; then
    sudo rm -f /etc/chrony/chrony.conf
else
    echo 'STOP: prior chrony.conf state is unknown'
    false
fi

sudo rm -f /etc/systemd/system/agibot_pm.service.d/20-clock-ordering.conf
sudo rm -f /etc/systemd/system/chrony.service.d/agibot-controls.conf
sudo rm -f /etc/systemd/system/agibot-clock-bootstrap.service
sudo rm -f /etc/chrony/sources.d/hope-approved.sources
sudo rm -f /etc/chrony/chrony-bootstrap.conf
sudo rm -f /etc/default/agibot-clock-bootstrap
```

Appendix A. Complete Deployment Artifacts

A.9 Metadata-preserving Rollback Commands (Part 2 of 2)

Generated from README section 11. Stop on any failed prerequisite or unknown prior state.

```
sudo rm -f /etc/default/agibot-timesync
sudo rm -f /usr/local/sbin/agibot-clock-bootstrap
sudo rm -f /usr/local/sbin/agibot-time-region
sudo rm -f /etc/agibot-time/current
sudo rm -f /etc/agibot-time/regions/china/chrony.sources
sudo rm -f /etc/agibot-time/regions/us/chrony.sources
sudo rm -f /etc/agibot-time/regions/europe/chrony.sources
sudo rm -f /var/lib/agibot-time/utc-unqualified-holdover
sudo rmdir /etc/agibot-time/regions/china 2>/dev/null || true
sudo rmdir /etc/agibot-time/regions/us 2>/dev/null || true
sudo rmdir /etc/agibot-time/regions/europe 2>/dev/null || true
sudo rmdir /etc/agibot-time/regions /etc/agibot-time 2>/dev/null || true
sudo rmdir /var/lib/agibot-time 2>/dev/null || true
sudo systemctl unmask chrony.service || true
sudo systemctl daemon-reload

BASE_CHRONY_ENABLED=$(head -n 1 "$BACKUP/chrony-enabled.txt")
case "$BASE_CHRONY_ENABLED" in
    enabled) sudo systemctl enable chrony.service ;;
    masked) sudo systemctl mask chrony.service ;;
    disabled|not-found|static|indirect|generated) ;;
    *) echo "Review unrecognized prior chrony enablement: $BASE_CHRONY_ENABLED" ;;
esac
sudo systemctl reset-failed

sudo sha256sum -c "$BACKUP/vendor-timesync.sha256"
sudo stat -Lc '%n %U:%G %a %s %y' \
    /opt/agibot/entry/system/timesync.sh
sudo cat "$BACKUP/vendor-timesync.stat"
sudo systemctl start agibot_roudi agibot_top agibot_ui
sudo systemctl start agibot_pm
pgrep -a -f 'ntp2rtc|chronyd|ptp4l|phc2sys' || true
```